

ECON1050 Tutorial 6

Single Variable Optimization

Semester 2 2026

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7 & 8 September

Today's Plan

- ▶ Recap: key ideas from Lecture 6
- ▶ Tutorial questions: Q1–Q5

Recap: Optimization and Global Extreme Points

Optimization

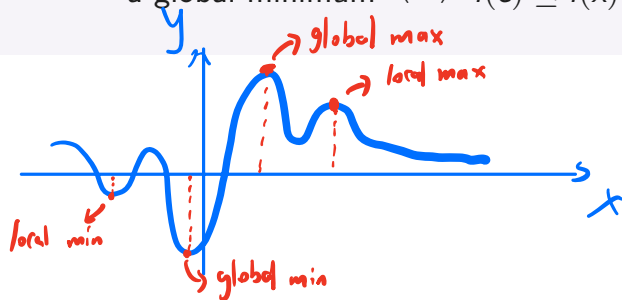
Optimization means finding the maximum or minimum of a univariate function over its domain.

Global extreme points

For a function f with domain D , a point $c \in D$ is

a global maximum $\iff f(c) \geq f(x) \quad \forall x \in D$,

a global minimum $\iff f(c) \leq f(x) \quad \forall x \in D$.



Recap: Stationary Points and the FOC

Stationary point

If f is differentiable, any interior maximum or minimum must occur at a point c where

$$f'(c) = 0.$$

Such points are called stationary or critical points.

First-order condition (FOC)

For an interior point c to be an extreme point of a differentiable function, the necessary condition is

$$f'(c) = 0.$$

The FOC is necessary, but by itself it is not sufficient to classify the point.

point is a optimal point $\Rightarrow f'(c) = 0$ stationary point

Recap: Extreme Value Theorem and Closed Intervals

Extreme Value Theorem

If f is continuous on a closed and bounded interval $[a, b]$, then f attains both a global maximum and a global minimum on $[a, b]$.

Procedure on $[a, b]$

1. Find all stationary points in (a, b) .
2. Evaluate f at every stationary point and at the endpoints a and b .
3. The largest value is the global maximum; the smallest is the global minimum.

Recap: Local vs Global Extreme Points

Local extreme point

A local maximum or minimum compares $f(c)$ only with values of $f(x)$ in some interval around c . A global extreme point compares $f(c)$ with all points in the domain.

Candidates for extreme points

Candidates include

- ▶ stationary points where $f'(x) = 0$,
- ▶ endpoints of the domain or interval,
- ▶ points where f is not differentiable.

First derivative test

$f'(x) : + \rightarrow - \implies c$ is a maximum,

$f'(x) : - \rightarrow + \implies c$ is a minimum.

If $f'(x)$ has the same sign on both sides of c , then c is not a local extreme point.

Recap: Second Derivative Test

Second-order conditions

At a stationary point c with $f'(c) = 0$,

$f''(c) < 0 \implies c$ is a local maximum,

$f''(c) > 0 \implies c$ is a local minimum,

$f''(c) = 0 \implies$ the test is inconclusive.

Concave and Convex Functions

On an interval I ,

$f''(x) > 0 \forall x \in I \implies f$ is strictly convex on I ,

$f''(x) < 0 \forall x \in I \implies f$ is strictly concave on I .

Extreme points under curvature

For an interior stationary point, concavity supports a maximum and convexity supports a minimum.

Recap: Inflection Points

Definition

An inflection point occurs where the curvature changes direction, that is, where $f''(x)$ changes sign.

Test

If f'' is continuous, an inflection point c must satisfy

$$f''(c) = 0.$$

However, $f''(c) = 0$ alone is not sufficient: the sign of $f''(x)$ must actually change across c .

Question 1

6 min

Find the global maximum and minimum of each function over the indicated interval.

1. $f(x) = x^3 - 3x + 8$, $x \in [-1, 2]$.
2. $f(x) = x^5 - 5x^3$, $x \in [-1, \sqrt{5}]$.

Tutorial Q1

Question 1

Find the global maximum and minimum of each function over the indicated interval.

1. $f(x) = x^3 - 3x + 8$, $x \in [-1, 2]$.
2. $f(x) = x^5 - 5x^3$, $x \in [-1, \sqrt{5}]$.

EVT

1. Find all stationary points in (a, b) .
2. Evaluate f at every stationary point and at the endpoints a and b .
3. The largest value is the global maximum; the smallest is the global minimum.

Find the global maximum and minimum of each function over the indicated interval.

1. $f(x) = x^3 - 3x + 8, \quad x \in [-1, 2]$.

2. $f(x) = x^5 - 5x^3, \quad x \in [-1, \sqrt{5}]$.

1. $f(x) = x^3 - 3x + 8, \quad x \in [-1, 2]$

$$f'(x) = 3x^2 - 3 = 0$$

$$3 \cdot (x^2 - 1) = 0$$

$$3 \cdot (x-1)(x+1) = 0$$

Stationary points $x = -1, 1$

$$f(-1) = (-1)^3 - 3 \cdot (-1) + 8 = 10 \quad \checkmark \Rightarrow \text{global max}$$

$$f(1) = 1^3 - 3 + 8 = 6 \quad \checkmark \Rightarrow \text{global min}$$

$$f(2) = 2^3 - 3 \cdot 2 + 8 = 10 \quad \checkmark \Rightarrow \text{global max}$$

2. $f(x) = x^5 - 5x^3, \quad x \in [-1, \sqrt{5}]$

$$f'(x) = 5x^4 - 15x^2 = 0$$

$$5x^2 \cdot (x^2 - 3) = 0$$

$$x = 0, -\sqrt{3}, \sqrt{3}$$

Stationary point $x = 0, \sqrt{3}$

$$f(0) = 0$$

$$f(\sqrt{3}) = -6\sqrt{3} \quad \text{lowest} \Rightarrow \text{global min}$$

$$f(-1) = 4 \quad \text{Highest} \Rightarrow \text{global max}$$

$$f(\sqrt{5}) = 0$$

Tutorial Q1(a): Global Maximum and Minimum

Knowledge used

- ▶ Extreme Value Theorem for a continuous function on a closed, bounded interval.
- ▶ FOC: find stationary points in the interior.
- ▶ Compare function values at stationary points and endpoints.

Solution

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1).$$

The roots are $x = -1$ and $x = 1$, but only $x = 1$ is an interior stationary point.

$$f(1) = 6, \quad f(-1) = 10, \quad f(2) = 10.$$

Therefore,

global maximum value 10 at $x = -1$ and $x = 2$,

global minimum value 6 at $x = 1$.

Tutorial Q1(b): Global Maximum and Minimum

Knowledge used

- ▶ Find all stationary points in the interior $(-1, \sqrt{5})$.
- ▶ Evaluate the function at the valid stationary points and the two endpoints.

Solution

$$f'(x) = 5x^4 - 15x^2 = 5x^2(x^2 - 3).$$

Stationary points are $x = 0, \pm\sqrt{3}$, but only 0 and $\sqrt{3}$ lie in $(-1, \sqrt{5})$.

$$f(0) = 0, \quad f(\sqrt{3}) = -6\sqrt{3},$$

$$f(-1) = 4, \quad f(\sqrt{5}) = 0.$$

Hence,

global maximum at $x = -1$, $f(-1) = 4$,

global minimum at $x = \sqrt{3}$, $f(\sqrt{3}) = -6\sqrt{3}$.

Tutorial Q2

Question 2

Find and classify the local extreme points for

$$f(x) = x^3 + 3x^2 - 2.$$

Tutorial Q2

Question 2

5 mins

Find and classify the local extreme points for

$$f(x) = x^3 + 3x^2 - 2.$$

Second-order conditions

At a stationary point c with $f'(c) = 0$,

$f''(c) < 0 \implies c$ is a local maximum,

$f''(c) > 0 \implies c$ is a local minimum,

$f''(c) = 0 \implies$ the test is inconclusive.

Find and classify the local extreme points for

$$f(x) = x^3 + 3x^2 - 2.$$

$$f(x) = x^3 + 3x^2 - 2$$

$$f'(x) = 3x^2 + 6x = 0$$

$$3x(x+2) = 0$$

stationary points $x = -2, 0$

$$f''(x) = 6x + 6$$

$$\Rightarrow f''(-2) = -2 \cdot 6 + 6 = -6 < 0 \text{ local maximum}$$

$$\Rightarrow f''(0) = 6 \cdot 0 + 6 = 6 > 0 \text{ local minimum}$$

Tutorial Q2: Stationary Points

Knowledge used

- FOC: interior candidates satisfy $f'(x) = 0$.

Solution

$$f'(x) = 3x^2 + 6x = 3x(x + 2).$$

Hence

$$f'(x) = 0 \iff 3x(x + 2) = 0,$$

so the stationary points are

$$x = -2 \quad \text{and} \quad x = 0.$$

Tutorial Q2: Classification

Knowledge used

- Second derivative test: $f''(c) < 0$ gives a local maximum; $f''(c) > 0$ gives a local minimum.

Solution

$$f''(x) = 6x + 6.$$

At $x = -2$,

$$f''(-2) = -6 < 0,$$

so $x = -2$ is a local maximum. At $x = 0$,

$$f''(0) = 6 > 0,$$

so $x = 0$ is a local minimum.

local max at $(-2, 2)$, local min at $(0, -2)$.

Question 3

Consider

$$f(x) = x^3 e^{-x}.$$

Find possible extreme points and inflection points.

Tutorial Q3

Question 3

7 mins

Consider

$$f(x) = x^3 e^{-x}.$$

Find possible extreme points and inflection points.

Test

If f' is continuous, an inflection point c must satisfy

$$f''(c) = 0.$$

However, $f''(c) = 0$ alone is not sufficient: the sign of $f''(x)$ must actually change across c .

Consider

$$f(x) = x^3 e^{-x}.$$

Find possible extreme points and inflection points.

$$f(x) = x^3 \cdot e^{-x}$$

$$f'(x) = 3x^2 \cdot e^{-x} + x^3 \cdot (-1) \cdot e^{-x} \Rightarrow f'(x) = e^{-x} \cdot x^2(3-x) = 0$$
$$= \underline{e^{-x} \cdot (3x^2 - x^3)} \quad x^2(3-x) = 0$$

Stationary points $x = 0, 3$

$$f''(x) = -e^{-x} \cdot (3x^2 - x^3) + e^{-x} \cdot (6x - 3x^2) \Rightarrow f''(x) = e^{-x} \cdot x(x^2 - 6x + 6) = 0$$
$$= e^{-x} \cdot (-3x^2 + x^3 + 6x - 3x^2)$$
$$= e^{-x} \cdot (x^3 - 6x^2 + 6x)$$
$$= e^{-x} \cdot x(x^2 - 6x + 6)$$
$$x(x^2 - 6x + 6) = 0$$
$$\downarrow \quad x^2 - 6x + 6 = 0$$
$$x = 0 \quad x = \frac{6 \pm \sqrt{36 - 24}}{2}$$
$$x = 3 \pm \sqrt{3}$$

Possible inflection points $x = 0, 3 - \sqrt{3}, 3 + \sqrt{3}$



$$\left. \begin{array}{l} f(-1) > 0 \\ f(1) < 0 \end{array} \right\} \Rightarrow x=0 \text{ is indeed an inflection point}$$

$$\left. \begin{array}{l} f(1) < 0 \\ f(3) > 0 \end{array} \right\} \Rightarrow x = 3 - \sqrt{3} \quad \checkmark$$

$$\left. \begin{array}{l} f(3) > 0 \\ f(5) < 0 \end{array} \right\} \Rightarrow x = 3 + \sqrt{3} \quad \checkmark$$

Tutorial Q3: First and Second Derivatives

Knowledge used

- Product rule and chain rule for derivatives of $x^3 e^{-x}$.

Solution

$$f'(x) = 3x^2 e^{-x} - x^3 e^{-x} = e^{-x}(3x^2 - x^3) = e^{-x}x^2(3 - x).$$

Differentiating again,

$$f''(x) = e^{-x}(6x - 6x^2 + x^3) = e^{-x}x(x^2 - 6x + 6).$$

Tutorial Q3: Possible Extreme Points

Knowledge used

- ▶ FOC to identify stationary points.
- ▶ Second derivative test when $f'(c) \neq 0$; if $f'(c) = 0$, use other information.

Solution

Since $e^{-x} > 0$,

$$f'(x) = 0 \iff x^2(3-x) = 0,$$

so the stationary points are $x = 0$ and $x = 3$. At $x = 3$,

$$f''(3) = -\frac{9}{e^3} < 0,$$

therefore

$x = 3$ is a local maximum.

At $x = 0$, $f'(0) = 0$, so the second derivative test is inconclusive.

Tutorial Q3: Inflection Points

Knowledge used

- ▶ Candidate inflection points satisfy $f''(x) = 0$.
- ▶ A genuine inflection point requires a sign change in $f''(x)$.

Solution

Since $e^{-x} > 0$,

$$f''(x) = 0 \iff x(x^2 - 6x + 6) = 0.$$

Hence

$$x = 0, \quad x = 3 - \sqrt{3}, \quad x = 3 + \sqrt{3}.$$

The sign diagram in the tutorial solution shows that $f''(x)$ changes sign at all three values. Therefore,

inflection points occur at $x = 0, 3 - \sqrt{3}, 3 + \sqrt{3}$.

Because $f'(x) > 0$ on both sides of $x = 0$, the stationary point $x = 0$ is not an extreme point.

Tutorial Q4

Question 4

5 min

A firm has

$$R(Q) = -0.0016Q^2 + 44Q$$

and

$$C(Q) = 0.0004Q^2 + 8Q + 64,000.$$

What production level Q maximises profit? Confirm that it is a maximum.

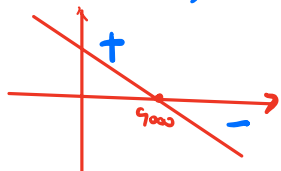
profit = Revenue - Cost

$$\begin{aligned}\pi(Q) &= R(Q) - C(Q) = -0.0016Q^2 + 44Q - 0.0004Q^2 - 8Q - 64,000 \\ &= -0.002Q^2 + 36Q - 64,000\end{aligned}$$



$$\pi'(Q) = -0.004Q + 36 = 0$$

Stationary point $Q = \frac{36}{0.004} = 9000$



$\pi'(Q) > 0$ when $Q < 9000$
 $\pi'(Q) \leq 0$ when $Q \geq 9000$ } $\Rightarrow Q = 9000$ is a ^{global} maximum point

Tutorial Q4: Profit Maximisation

Knowledge used

- ▶ Profit is revenue minus cost: $\Pi(Q) = R(Q) - C(Q)$.
- ▶ FOC: $\Pi'(Q) = 0$ identifies the candidate optimum.
- ▶ SOC/concavity confirms whether the candidate is a maximum.

Solution

$$\Pi(Q) = R(Q) - C(Q) = -0.002Q^2 + 36Q - 64,000.$$

FOC:

$$\Pi'(Q) = -0.004Q + 36 = 0 \quad \Rightarrow \quad \boxed{Q^* = 9000}.$$

SOC:

$$\Pi''(Q) = -0.004 < 0.$$

Thus $\Pi(Q)$ is strictly concave, so

$$\boxed{Q = 9000 \text{ is the profit-maximising production level}}.$$

Question 5

5 min

Given

$$f(x) = \frac{e^x}{2 + e^{2x}},$$

find the intervals in which the function is increasing and decreasing. Find the minimum and maximum points.

Question 5

5 min

Given

$$f(x) = \frac{e^x}{2 + e^{2x}},$$

find the intervals in which the function is increasing and decreasing. Find the minimum and maximum points.

First derivative test

$f'(x) : + \rightarrow - \implies c$ is a maximum,

$f'(x) : - \rightarrow + \implies c$ is a minimum.

If $f'(x)$ has the same sign on both sides of c , then c is not a local extreme point.

Given

$$f(x) = \frac{e^x}{2 + e^{2x}},$$

find the intervals in which the function is increasing and decreasing. Find the minimum and maximum points.

$$f(x) = \frac{e^x}{2 + e^{2x}}$$

$$f'(x) = \frac{e^x \cdot (2 + e^{2x}) - e^x \cdot 2 \cdot e^{2x}}{(2 + e^{2x})^2}$$

$$= \frac{e^x \cdot (2 + e^{2x} - 2 \cdot e^{2x})}{(2 + e^{2x})^2}$$

$$= \frac{e^x \cdot (2 - e^{2x})}{(2 + e^{2x})^2}$$

$$2 - e^{2x} = 0$$

$$2 = e^{2x}$$

$$\ln 2 = \ln e^{2x}$$

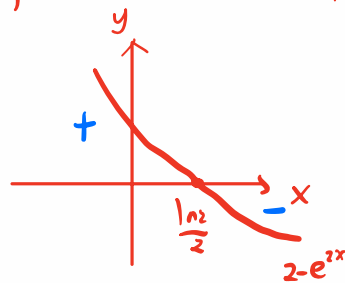
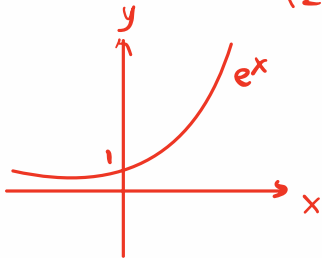
$$\ln 2 = 2x$$

$$x = \frac{\ln 2}{2}$$

$$f'(x) = \frac{e^x(2 - e^{2x})}{(2 + e^{2x})^2} = 0$$

$$2 - e^{2x} = 0$$

$$\text{Stationary point } x = \frac{\ln 2}{2}$$

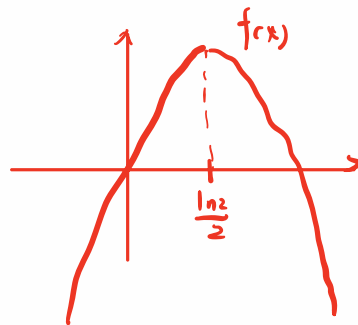


$\Rightarrow x < \frac{\ln 2}{2}, f'(x) > 0 \Rightarrow f(x)$ is increasing

$x > \frac{\ln 2}{2}, f'(x) < 0 \Rightarrow f(x)$ is decreasing

$\Rightarrow$

This is exactly the first derivative test
the stationary point $x = \frac{\ln 2}{2}$ is a max point



$\Rightarrow$ no global min or local min

Tutorial Q5: Derivative and Critical Point

Knowledge used

- ▶ Quotient rule to compute $f'(x)$.
- ▶ The denominator is always positive, so the numerator determines the sign of $f'(x)$.

Solution

$$f'(x) = \frac{e^x(2 + e^{2x}) - e^x(2e^{2x})}{(2 + e^{2x})^2} = \frac{e^x(2 - e^{2x})}{(2 + e^{2x})^2}.$$

Since $e^x > 0$ and the denominator is positive,

$$f'(x) = 0 \iff 2 - e^{2x} = 0 \iff 2x = \ln 2.$$

Thus the only critical point is

$$x = \frac{\ln 2}{2}.$$

Tutorial Q5: Increasing and Decreasing Intervals

Knowledge used

- Use the sign of the first derivative to determine monotonicity.

Solution

For $x < \frac{\ln 2}{2}$, we have $e^{2x} < 2$, so $f'(x) > 0$. For $x > \frac{\ln 2}{2}$, we have $e^{2x} > 2$, so $f'(x) < 0$. Therefore,

f is increasing on $\left(-\infty, \frac{\ln 2}{2}\right]$

and

f is decreasing on $\left[\frac{\ln 2}{2}, \infty\right)$.

Tutorial Q5: Extreme Points

Knowledge used

- ▶ First derivative test: increasing then decreasing implies a maximum.
- ▶ On an unbounded domain, a global minimum need not be attained.

Solution

Because f increases to the left of $\frac{\ln 2}{2}$ and decreases to the right,

$$x = \frac{\ln 2}{2} \text{ is the global maximum point.}$$

Its value is

$$f\left(\frac{\ln 2}{2}\right) = \frac{\sqrt{2}}{4}.$$

Also,

$$f(x) > 0 \quad \text{for all } x, \quad \lim_{x \rightarrow \pm\infty} f(x) = 0.$$

Hence the infimum is 0, but it is not attained; therefore there is no global minimum point.

Key Takeaways

- ▶ Interior candidates for extrema satisfy the FOC $f'(c) = 0$, but classification requires additional information.
- ▶ On a closed interval, compare all interior stationary points with the endpoints to find global extrema.
- ▶ The first derivative test and second derivative test classify local maxima and minima.
- ▶ Inflection points require a change in the sign of $f''(x)$; $f''(c) = 0$ alone is not enough.
- ▶ Concavity and convexity provide powerful global information for optimisation problems.